M.A.M. SCHOOL OF ENGINEERING

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Affiliated to Anna University, Chennai
Siruganur, Trichy -621 105. | www.mamse.in

Department of Aeronautical Engineering
(Common to EEE,Mech,Mecht)
Academic Year 2025-2026 (Even Semester)
CIA – I Exam

Unit - I

Sub.Code/Sub.Name : OCS351- IOT Concepts and Applications
Year/Sem. : III Year/VI Sem
Date :
Duration : 3 Hrs
Max. Marks : 100

	Part A (10 X 2 = 20 Marks) Answer All the Questions	**M.A**	**CO**	**BTL**
1.	Define the Internet of Things (IoT).	2	CO1	K2
2.	What are two key enabling technologies of IoT?	2	CO1	K2
3.	List any two layers of the Core IoT Functional Stack.	2	CO1	K3
4.	What is the role of Fog computing in IoT?	2	CO1	K2
5.	Differentiate between Edge computing and Cloud computing in IoT.	2	CO1	K2
6.	What is the main objective of oneM2M architecture?	2	CO1	K2
7.	Name two alternative IoT models apart from oneM2M and IoT World Forum.	2	CO1	K1
8.	What is a major advantage of using the IoT World Forum (IoTWF) architecture?	2	CO1	K2
9.	Mention two benefits of using Cloud computing in IoT.	2	CO1	K2
10.	What is the significance of a Simplified IoT Architecture?	2	CO1	K1

		Part B	**M.A**	**CO**	**BTL**
11.	a)	Explain the evolution of IoT, covering its growth from basic machine-to-machine (M2M) communication to modern-day IoT applications.	8	CO1	K1
12.	a)	Discuss the enabling technologies of IoT, such as sensors, communication protocols, cloud computing, and artificial intelligence.	16	CO1	K2
13.	a)	Compare and contrast the oneM2M and IoT World Forum (IoTWF) architectures. Highlight their structure, advantages, and use cases.	16	CO1	K3
14.	a)	Describe the role of Fog, Edge, and Cloud computing in IoT. How do they contribute to efficient IoT implementations? Provide examples.	16	CO1	K2
15.	a)	Illustrate the Core IoT Functional Stack and explain its different layers. How does it simplify IoT architecture and functionality?	16	CO1	K3

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Unit - II

Department of Aeronautical Engineering
(Common to EEE, Mech, Mecht)
Academic Year 2025-2026 (Even Semester)
CIA – I Exam

Sub.Code/Sub.Name	: OCS351- IOT Concepts and Applications	**Date**	:
Year/Sem.	: III Year/VI Sem	**Duration**	: 3 Hrs
		Max. Marks	: 100

	Part A (10 X 2 = 20 Marks) **Answer All the Questions**	**M.A**	**CO**	**BTL**
1.	What are the functional blocks of an IoT ecosystem?	2	CO1	K1
2.	Define a sensor in the context of IoT.	2	CO1	K1
3.	Give two examples of actuators used in IoT applications.	2	CO1	K1
4.	What is the role of smart objects in an IoT system?	2	CO1	K2
5.	Name two common communication modules used in IoT.	2	CO1	K2

		Part B	**M.A**	**CO**	**BTL**
11.	a)	Explain the functional blocks of an IoT ecosystem. Discuss their roles and importance in building a complete IoT system.	16	CO1	K2
12.	a)	Illustrate different types of sensors & actuators with real-time examples	16	CO1	K3
13.	a)	Discuss different communication modules used in IoT (Bluetooth, Zigbee). Compare their features, advantages, and applications.	16	CO1	K3

Department of AERO,MECH,MCT,EEE

Unit - II

Academic Year 2025-2026 (Even Semester)

CIA – II Exam

Sub.Code/Sub.Name	OCS352- IOT Concepts And Applications	**Date**	
Year/Sem.	03 / 06	**Duration**	3 Hrs
		Max. Marks	100

Sl. No.	**Part A (10 X 2 = 20 Marks)** **Answer All the Questions**	**M.A**	**CO**	**BTL**
1.	Differentiate between Bluetooth and Zigbee in IoT communication.	2	CO2	K2
2.	What is the function of a control unit in an IoT system?	2	CO2	K2
3.	Mention two advantages of using Wi-Fi in IoT applications.	2	CO2	K1
4.	How does GPS contribute to IoT applications?	2	CO2	K2
5.	What is the primary purpose of GSM modules in an IoT system?	2	CO2	K2

Sl. No.	**Part B**	**M.A**	**CO**	**BTL**
6.	Analyze the role of control units in an IoT system. How do they process data from sensors and control actuators? Give real-world examples.	13	CO2	K3
7.	Explain the architecture and working principle of Wi-Fi technology. Discuss the different IEEE 802.11 standards and compare their features. Analyze the advantages and limitations of Wi-Fi in modern communication systems.	13	CO2	K2
8.	Discuss the architecture and functioning of GPS. Evaluate the various sources of errors affecting GPS accuracy and suggest methods to improve positioning accuracy.	13	CO2	K3

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Unit - III

Academic Year 2025-2026 (Even Semester)

CIA – II Exam

Sub.Code/Sub.Name	OCS352- IOT Concepts And Applications	**Date**	
Year/Sem.	03 / 06	**Duration**	3 Hrs
		Max. Marks	100

Sl. No.	Part A (10 X 2 = 20 Marks) Answer All the Questions	M.A	CO	BTL
1.	What is the role of IPv6 in IoT?	2	CO3	K1
2.	Expand 6LoWPAN and explain its significance in IoT.	2	CO3	K1
3.	How does MQTT facilitate communication in IoT applications?	2	CO3	K2
4.	What are the key features of CoAP in IoT environments?	2	CO3	K2
5.	Differentiate between MQTT and CoAP in terms of communication model and use cases.	2	CO3	K2
6.	Define a Wireless Sensor Network (WSN).	2	CO3	K1
7.	Differentiate between public and private cloud.	2	CO3	K1
8.	List two use cases of Big Data Analytics in the healthcare sector.	2	CO3	K2
9.	Illustrate the role of an RTOS in Embedded Systems.	2	CO3	K2
10.	Identify two challenges in data collection from sensor nodes in WSNs.	2	CO3	K2

Sl. No.	Part B	M.A	CO	BTL
11.	Explain the fundamental concepts of IPv6 and CoAP. Describe their roles in IoT communication and networking.	16	CO3	K2
12.	Discuss the working principles of 6LoWPAN and MQTT in IoT. How do they enhance communication efficiency in constrained environments?	16	CO3	K3
13.	Discuss the architecture of a typical Wireless Sensor Network. Illustrate with a case study or application.	16	CO3	K2
14.	Compare and contrast Cloud Computing and Big Data Analytics with respect to architecture, data handling, and scalability. Evaluate the benefits and limitations of integrating both technologies for enterprise-level solutions.	16	CO3	K3
15.	Explain the working principle of an RFID system with a neat diagram. Describe the components of RFID such as tag, reader, and antenna. Discuss the types of RFID tags (active, passive, semi-passive) with suitable examples.	16	CO3	K2

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Unit - V

Department of AERO,MECH,MCT,EEE

Academic Year 2025-2026 (Even Semester)

CIA – III Exam

Sub.Code/Sub.Name	OCS352- IOT Concepts And Applications	**Date**	
Year/Sem.	03 / 06	**Duration**	3 Hrs
		Max. Marks	100

Sl. No.	Part A (10 X 2 = 20 Marks) Answer All the Questions	M.A	CO	BTL
1.	Define Industrial IoT (IIoT).	2	CO5	K2
2.	List any two applications of IoT in smart cities.	2	CO5	K2
3.	What is meant by home automation in IoT?	2	CO5	K3
4.	Mention any two IoT-enabled smart health devices.	2	CO5	K2
5.	Define the term "Smart Mobility".	2	CO5	K2
6.	Name two sensors commonly used in environmental monitoring systems.	2	CO5	K2
7.	What is a vertical business model in IoT?	2	CO5	K1
8.	State one advantage of IoT in precision agriculture.	2	CO5	K2
9.	List any two communication technologies used in smart transportation systems.	2	CO5	K2
10.	What is the role of surveillance systems in smart cities?	2	CO5	K1

Sl. No.	Part B	M.A	CO	BTL
11.	Explain the concept of a Smart City. Describe the key components and how IoT is used to improve city infrastructure and public services.	16	CO5	K1
12.	Define Smart Health. List and explain four IoT applications in healthcare systems.	16	CO5	K2
13.	Discuss how IoT is applied in Smart Agriculture. Explain the roles of sensors, automation, and data analytics in improving crop yield.	16	CO5	K3
14.	Describe different business models used for Internet of Things applications. Provide examples for each model.	16	CO5	K2
15.	Define Smart Mobility. Describe how IoT enables efficient public transportation and traffic management systems.	16	CO5	K3

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Unit - IV

Academic Year 2025-2026 (Even Semester)

CIA – III Exam

Sub.Code/Sub.Name	OCS352- IOT Concepts And Applications	**Date**	
Year/Sem.	03 / 06	**Duration**	3 Hrs
		Max. Marks	100

Sl. No.	**Part A (10 X 2 = 20 Marks)** **Answer All the Questions**	**M.A**	**CO**	**BTL**
1.	Define GPIO in the context of Raspberry Pi	2	CO4	K1
2.	What is the role of a microcontroller in IoT applications?	2	CO4	K1
3.	List two programming languages commonly used with Raspberry Pi for IoT development.	2	CO4	K1
4.	Mention any two cloud platforms used in IoT deployment.	2	CO4	K2
5.	State one difference between digital and analog signals in IoT interfacing.	2	CO4	K2
6.	Write a Python command to set a GPIO pin as an output in Raspberry Pi.	2	CO4	K2
7.	How would you connect a DHT11 sensor to an Arduino board?	2	CO4	K2
8.	Explain the term ‘pinMode()’ in Arduino programming.	2	CO4	K1
9.	What happens if GPIO pin direction (input/output) is not configured properly?	2	CO4	K2
10.	Mention two advantages of using Raspberry Pi over Arduino for cloud-connected IoT systems.	2	CO4	K2

Sl. No.	**Part B**	**M.A**	**CO**	**BTL**
11.	Define Internet of Things (IoT). Explain the key components of an IoT system and list common hardware platforms used for IoT deployments.	16	CO4	K2
12.	Explain the architecture of Raspberry Pi and its suitability for IoT applications. Describe its key hardware features and available communication interfaces.	16	CO4	K2
13.	List and explain the functions of five commonly used GPIO pins on Raspberry Pi. Also explain the difference between input and output mode in GPIO.	16	CO4	K3
14.	Describe the basic programming structure used in Arduino IDE for IoT applications. Explain the functions setup() and loop() with suitable examples.	16	CO4	K3
15.	Explain how an IoT device connects to the cloud. Describe common communication protocols used and mention any two cloud platforms with their features.	16	CO4	K2